

```
"""
STEP 0: Install Required Libraries
```

This cell installs spaCy and downloads the English language model.  
spaCy is used for Natural Language Processing tasks like tokenization,  
POS tagging, and dependency parsing.

```
"""

# Install spaCy
!pip install spacy

# Download English model
!python -m spacy download en_core_web_sm
```

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Requirement already satisfied: spacy in /usr/local/lib/python3.12/dist-packages (3.8.11)
Requirement already satisfied: spacy-legacy<3.1.0,>=3.0.11 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.0.12)
Requirement already satisfied: spacy-loggers<2.0.0,>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.5)
Requirement already satisfied: murmurhash<1.1.0,>=0.28.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.15)
Requirement already satisfied: cytoolz<2.1.0,>=2.0.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.13)
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Requirement already satisfied: thinc<8.4.0,>=8.3.4 in /usr/local/lib/python3.12/dist-packages (from spacy) (8.3.10)
Requirement already satisfied: wasabi<1.2.0,>=0.9.1 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.1.3)
Requirement already satisfied: srsly<3.0.0,>=2.4.3 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.5.2)
Requirement already satisfied: catalogue<2.1.0,>=2.0.6 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.10)
Requirement already satisfied: weasel<0.5.0,>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.4.3)
Requirement already satisfied: typer-slim<1.0.0,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.24.0)
Requirement already satisfied: tqdm<5.0.0,>=4.38.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (4.67.3)
Requirement already satisfied: numpy>=1.19.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.2)
Requirement already satisfied: requests<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.32.4)
Requirement already satisfied: pydantic!=1.8,!1.8.1,<3.0.0,>=1.7.4 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.11.10)
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Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (26.0)
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!1.8.1,<3.0.0,>=1.7.4) (0.7.0)
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!1.8.1,<3.0.0,>=1.7.4) (2.41.4)
Requirement already satisfied: typing-extensions>=4.14.1 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!1.8.1,<3.0.0,>=1.7.4) (4.14.1)
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!1.8.1,<3.0.0,>=1.7.4) (0.4.2)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0) (3.4.0)
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Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0) (2025.11.12)
Requirement already satisfied: blis<1.4.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4->spacy) (1.3.0)
Requirement already satisfied: confection<1.0.0,>=0.0.1 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4->spacy) (0.0.1)
Requirement already satisfied: typer>=0.24.0 in /usr/local/lib/python3.12/dist-packages (from typer-slim<1.0.0,>=0.3.0->spacy) (0.24.0)
Requirement already satisfied: cloudpathlib<1.0.0,>=0.7.0 in /usr/local/lib/python3.12/dist-packages (from weasel<0.5.0,>=0.4.2->spacy) (0.7.0)
Requirement already satisfied: smart-open<8.0.0,>=5.2.1 in /usr/local/lib/python3.12/dist-packages (from weasel<0.5.0,>=0.4.2->spacy) (7.0.5)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from Jinja2->spacy) (3.0.3)
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart-open<8.0.0,>=5.2.1->weasel<0.5.0,>=0.4.2->spacy) (1.17.0)
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-slim<1.0.0,>=0.3.0) (8.2.1)
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-slim<1.0.0,>=0.3.0) (1.5.4)
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-slim<1.0.0,>=0.3.0) (13.9.0)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-slim<1.0.0,>=0.3.0) (0.0.2)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer>=0.24.0) (3.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer>=0.24.0) (2.19.2)
Requirement already satisfied: mdurl~>0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3.0->typer>=0.24.0) (0.1.2)
Collecting en-core-web-sm==3.8.0
  Downloading https://github.com/explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en_core_web_sm-3.8.0-py3-none-any-12.8/12.8 MB 84.8 MB/s eta 0:00:00
```

✓ Download and installation successful

You can now load the package via `spacy.load('en_core_web_sm')`

⚠ Restart to reload dependencies

If you are in a Jupyter or Colab notebook, you may need to restart Python in order to load all the package's dependencies. You can do this by selecting the 'Restart kernel' or 'Restart runtime' option.

```
"""
STEP 1: Load spaCy Language Model
```

Language models contain statistical data required for NLP tasks such as:

- Tokenization
- POS tagging
- Dependency parsing

'en\_core\_web\_sm' is a small English model suitable for basic NLP tasks.

```
"""

# Import spaCy
import spacy
```

```
# Load English model
nlp = spacy.load("en_core_web_sm")

# Verify model loading
print("spaCy model loaded successfully!")
```

spaCy model loaded successfully!

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STEP 2: Prepare Dataset

We define at least 15 sentences of different types:

- Simple
- Compound
- Complex
- Interrogative
- Passive voice

These sentences simulate real-world data such as news, social media, and technical text.

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```
sentences = [
    "The cat sits on the mat.",
    "She is reading a book.",
    "John and Mary went to the market.",
    "Although it was raining, we went outside.",
    "He completed the assignment and submitted it.",
    "The ball was thrown by the boy.",
    "Can you solve this problem?",
    "Artificial intelligence is transforming industries.",
    "The teacher explained the concept clearly.",
    "After the meeting ended, everyone left quickly.",
    "The car was repaired by the mechanic.",
    "She sings beautifully.",
    "Do you like programming?",
    "The scientist who discovered the formula won a prize.",
    "They are playing football in the ground."
]

print("Total sentences:", len(sentences))
```

Total sentences: 15

```
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STEP 3: Sentence Processing using spaCy

spaCy processes text using an NLP pipeline:

1. Tokenization → Splits text into words
2. POS Tagging → Assigns grammatical tags
3. Dependency Parsing → Identifies relationships between words

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```
# Process all sentences
docs = [nlp(sentence) for sentence in sentences]

print("Sentences processed successfully!")
```

Sentences processed successfully!

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STEP 4: Extract Dependency Relations

For each token in each sentence, we extract:

- Token text
- Lemma (base form)
- POS tag
- Dependency label
- Head word (parent in tree)

These help us understand grammatical structure.

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```
import pandas as pd

# Store all results
```

```

all_data = []

# Loop through sentences
for doc in docs:
    for token in doc:
        all_data.append([
            doc.text,
            token.text,
            token.lemma_,
            token.pos_,
            token.dep_,
            token.head.text
        ])

# Create DataFrame
df = pd.DataFrame(all_data, columns=[
    "Sentence", "Token", "Lemma", "POS", "Dependency", "Head"
])

# Display first 20 rows
df.head(20)

```

|    | Sentence                          | Token   | Lemma  | POS   | Dependency | Head    |
|----|-----------------------------------|---------|--------|-------|------------|---------|
| 0  | The cat sits on the mat.          | The     | the    | DET   | det        | cat     |
| 1  | The cat sits on the mat.          | cat     | cat    | NOUN  | nsubj      | sits    |
| 2  | The cat sits on the mat.          | sits    | sit    | VERB  | ROOT       | sits    |
| 3  | The cat sits on the mat.          | on      | on     | ADP   | prep       | sits    |
| 4  | The cat sits on the mat.          | the     | the    | DET   | det        | mat     |
| 5  | The cat sits on the mat.          | mat     | mat    | NOUN  | pobj       | on      |
| 6  | The cat sits on the mat.          | .       | .      | PUNCT | punct      | sits    |
| 7  | She is reading a book.            | She     | she    | PRON  | nsubj      | reading |
| 8  | She is reading a book.            | is      | be     | AUX   | aux        | reading |
| 9  | She is reading a book.            | reading | read   | VERB  | ROOT       | reading |
| 10 | She is reading a book.            | a       | a      | DET   | det        | book    |
| 11 | She is reading a book.            | book    | book   | NOUN  | dobj       | reading |
| 12 | She is reading a book.            | .       | .      | PUNCT | punct      | reading |
| 13 | John and Mary went to the market. | John    | John   | PROPN | nsubj      | went    |
| 14 | John and Mary went to the market. | and     | and    | CCONJ | cc         | John    |
| 15 | John and Mary went to the market. | Mary    | Mary   | PROPN | conj       | John    |
| 16 | John and Mary went to the market. | went    | go     | VERB  | ROOT       | went    |
| 17 | John and Mary went to the market. | to      | to     | ADP   | prep       | went    |
| 18 | John and Mary went to the market. | the     | the    | DET   | det        | market  |
| 19 | John and Mary went to the market. | market  | market | NOUN  | pobj       | to      |

```

"""
STEP 5: Identify Sentence Structure

We extract:
- Subject (nsubj)
- Object (dobj / pobj)
- ROOT verb
- Modifiers (amod, advmod)

Dependency parsing helps identify relationships like:
who did what to whom.
"""

for doc in docs:
    print("\nSentence:", doc.text)

    subject = []
    obj = []
    root = ""

```

```

modifiers = []

for token in doc:
    if token.dep_ == "nsubj":
        subject.append(token.text)
    if token.dep_ in ["dobj", "pobj"]:
        obj.append(token.text)
    if token.dep_ == "ROOT":
        root = token.text
    if token.dep_ in ["amod", "advmod"]:
        modifiers.append(token.text)

print("Subject:", subject)
print("Verb (ROOT):", root)
print("Object:", obj)
print("Modifiers:", modifiers)

```

```

Subject: []
Verb (ROOT): thrown
Object: ['boy']
Modifiers: []

```

```

Sentence: Can you solve this problem?
Subject: ['you']
Verb (ROOT): solve
Object: ['problem']
Modifiers: []

```

```

Sentence: Artificial intelligence is transforming industries.
Subject: ['intelligence']
Verb (ROOT): transforming
Object: ['industries']
Modifiers: ['Artificial']

```

```

Sentence: The teacher explained the concept clearly.
Subject: ['teacher']
Verb (ROOT): explained
Object: ['concept']
Modifiers: ['clearly']

```

```

Sentence: After the meeting ended, everyone left quickly.
Subject: ['meeting', 'everyone']
Verb (ROOT): left
Object: []
Modifiers: ['quickly']

```

```

Sentence: The car was repaired by the mechanic.
Subject: []
Verb (ROOT): repaired
Object: ['mechanic']
Modifiers: []

```

```

Sentence: She sings beautifully.
Subject: ['She']
Verb (ROOT): sings
Object: []
Modifiers: ['beautifully']

```

```

Sentence: Do you like programming?
Subject: ['you']
Verb (ROOT): like
Object: ['programming']
Modifiers: []

```

```

Sentence: The scientist who discovered the formula won a prize.
Subject: ['scientist', 'who']
Verb (ROOT): won
Object: ['formula', 'prize']
Modifiers: []

```

```

Sentence: They are playing football in the ground.
Subject: ['They']
Verb (ROOT): playing
Object: ['football', 'ground']
Modifiers: []

```

"""

#### STEP 6: Dependency Tree Visualization

displaCy is spaCy's built-in visualization tool.  
It graphically represents dependency relationships between words.

This helps in understanding sentence structure visually.  
"""

```
from spacy import displacy
```

```
# Visualize first 5 sentences
```

```
for i in range(5):
```

```
    doc = docs[i]
```

```
    displacy.render(doc, style="dep", jupyter=True)
```

